

deployment hardware.

The avionics bay serves as the structural and functional core of the vehicle. It houses the flight computer, inertial measurement unit (IMU), power regulation hardware, and wiring harness. The avionics were enclosed within a protective internal sleeve to shield sensitive electronics from combustion byproducts produced during black powder motor operation, including carbon dioxide, nitrogen, and carbon monoxide. This enclosure also provided structural isolation from exhaust turbulence and vibration.

The thrust assembly consists of the motor mount, retention system, and exhaust shielding designed to interface directly with the TVC gimbal. The motor mount was 3D printed with high infill density to maximize rigidity while minimizing mass and preventing unwanted shifts in the center of mass. An Estes E12-4 solid rocket motor was selected for this prototype due to its predictable thrust curve and compatibility with the airframe scale. The motor mount incorporates mechanical retention screws positioned beneath the solid propellant grain to prevent axial movement during ignition and powered flight. When the gimbal is at its neutral position, the motor centerline is aligned with the rocket's longitudinal axis to avoid introducing unintended moments.

Passive aerodynamic fins were intentionally omitted from the design. This decision allowed thrust vector control to serve as the sole stabilization mechanism, more closely modeling real-world TVC applications such as the Falcon 9 first stage. While active or passive fins can provide stabilization at higher airspeeds, their exclusion ensured that all attitude correction authority originated from the gimbal system alone.

2.2 Thrust Vectoring (TVC) System Design

The active stabilization system is centered around a two-axis thrust vector control gimbal that allows the rocket motor to deflect independently in pitch and yaw relative to the vehicle's longitudinal axis. This capability enables real-time corrective torque application during powered flight.

The gimbal mechanism consists of a nested, dual-axis structure mounted within the aft section of the airframe. It was designed as a multi-part PLA assembly to maximize available degrees of freedom while maintaining sufficient stiffness under thrust loading. Two MG90S digital micro servos actuate the gimbal, selected for their favorable torque-to-weight ratio, rapid response time (approximately 0.12 seconds per 60 degrees), and ease of integration with the flight computer.

The gimbal provides a maximum deflection range of ± 10 degrees in both pitch and yaw. This range was determined through simulation and mechanical testing to provide sufficient corrective authority for a rocket of this size while avoiding excessive coupling effects or mechanical interference. Because corrective torque is directly proportional to the deflection angle, this limit defines the upper bound of achievable control authority during powered flight.

2.3 Avionics & Sensor Suite

Vehicle orientation and acceleration are measured using an MPU-6050 inertial measurement unit, which integrates a three-axis gyroscope and three-axis accelerometer into a single module. The gyroscope provides high-rate angular velocity measurements suitable for short-term attitude estimation, while the accelerometer provides absolute orientation reference under low-dynamic conditions. Sensor data is transmitted digitally to the flight computer via the I²C protocol. Although the MPU-6050 lacks onboard magnetometer or barometric sensors, its high update rate, affordability, and compatibility with embedded development platforms made it suitable for this prototype. Its limitations are addressed through software-based sensor fusion, described in Section 3.2. Flight data is logged using the Teensy 4.1’s onboard SD card interface at frequencies between 100 and 200 Hz. Logged parameters include IMU measurements, gimbal deflection angles, system state transitions, and apogee detection events. This high temporal resolution enables detailed post-flight analysis of control system performance and disturbance response. Several onboard LEDs provide real-time visual feedback of system status, including power availability, IMU initialization, and flight mode. These indicators proved useful during ground testing and pre-launch diagnostics. Power is supplied by a two-cell lithium polymer battery regulated to 5 V via a buck converter. The system is designed to handle transient current spikes caused by servo actuation, with additional capacitive smoothing included to mitigate voltage sag. Modular JST and Dupont connectors were used throughout the wiring harness to facilitate rapid component replacement and reduce the risk of permanent damage during testing.

2.4 Onboard Flight Computer & Software Stack

The flight computer is built around a Teensy 4.1 microcontroller, selected for its high computational throughput, reliability, and integrated storage capabilities. Its ARM Cortex-M7 processor operates at 600 MHz, providing ample processing headroom for real-time control loops, Kalman filtering, and state machine logic without risk of timing overruns. The board’s extensive I/O support allows simultaneous interfacing with the IMU, servos, status indicators, and safety switches.

The software architecture is organized as a hierarchical state machine, detailed further in Section 4. At the highest level, the system transitions between predefined operational modes including standby, countdown, powered flight, apogee detection, descent, and recovery. Within each state, dedicated routines handle sensor acquisition, PID computation, actuator commands, and data logging.

The flight software is implemented using Arduino-based C++, leveraging standard libraries for I²C communication, SD card access, and servo control, alongside a custom Kalman filter implementation. This development approach combines the accessibility of the Arduino ecosystem with the computational performance required for advanced control and data analysis.

2.5 Justification for TVC vs. Passive Fins

Compared to passive aerodynamic stabilization methods, thrust vector control offers significantly greater authority during low-speed and early ascent phases, where aerodynamic forces are minimal.

Static fins rely on airflow to generate stabilizing moments and therefore become effective only after sufficient velocity has been achieved. In contrast, thrust vectoring allows corrective forces to be applied immediately after ignition, when disturbances and asymmetries are most likely to grow rapidly.

This capability is particularly important for low-mass rockets, which are highly sensitive to wind, thrust misalignment, and manufacturing tolerances. Small deviations in attitude can quickly compound into large trajectory errors if not corrected early. By actively redirecting thrust, the TVC system provides precise and anticipatory corrections, reducing reliance on passive aerodynamic damping.

Furthermore, the omission of fins results in a vehicle that is inherently aerodynamically unstable, as the center of pressure (*CoP*) often remains forward of the center of mass (*CoM*) during certain flight regimes. This configuration effectively mimics the 'inverted pendulum' challenge found in orbital-class boosters, necessitating the active, high-frequency balancing performance that only a TVC system can provide.

2.6 System Integration Considerations and Challenges

Integrating mechanical, electrical, and computational subsystems into a cohesive thrust-vector-controlled vehicle presents significant engineering challenges. The reliability of the rocket depends not only on the performance of individual components but also on their interaction under conditions of high acceleration, vibration, and thermal stress.

Thermal protection was a primary concern due to the proximity of avionics to the solid rocket motor. Exhaust gases and particulate matter generated during combustion pose a risk to electronic components. To mitigate this, the motor projection was positioned to direct exhaust away from sensitive structures, and a lightweight 3D-printed PLA shield wrapped in aluminum foil was installed to provide thermal isolation.

Vibration posed another significant challenge, particularly for inertial sensors. High-frequency oscillations can introduce noise into IMU measurements and degrade attitude estimation. To address this, the IMU was mounted using rubber isolation to attenuate vibration, while the gimbal mount and avionics bay were reinforced with cross-braced supports to reduce torsional flexing. Wiring harnesses were secured using adhesive and strain relief to prevent movement during launch.

Maintaining alignment between the center of mass (*CoM*) and center of thrust (*CoT*) is critical in thrust-vector-controlled systems. Even small offsets can introduce unwanted coupling effects that reduce control effectiveness. The *CoM* was determined experimentally using a balance method with the fully assembled rocket, including the motor. This measurement was supplemented with a method-of-moments calculation based on individual component masses and positions to increase confidence in the result. The equation for calculating the moments was derived from the *CoM* equation:

$$x_{CoM} = \frac{\sum x_i m_i}{\sum m_i}$$

Therefore, the total center of mass was determined from:

$$x_{cg} = \frac{\sum x_i W_i}{\sum W_i}$$

Where W_i represents the component's weight and x_i its distance from the fixed point. This method allowed for high confidence in the CoM measurement by combining balance testing with quantitative modeling.

The center of pressure (CoP) was estimated using OpenRocket simulation software, which applies the Barrowman equations to model aerodynamic forces on slender bodies. The final configuration maintained the CoM approximately 80 mm ahead of the CoP, providing a favorable static margin. Additionally, propellant burn during powered flight causes the CoM to shift forward, further improving stability during coast. Center of Pressure is governed by the equation:

$$x_{CoP} = \frac{\sum C_{N_i} x_i}{\sum C_{N_i}}$$

Where C_{N_i} is the normal force coefficient for each section, and x_i represents its axial distance from the nose tip. OpenRocket's output provided both a numerical CoP location and an estimated margin given by:

$$\text{Static Margin} = \frac{(x_{CoM} - x_{CoP})}{D}$$

Where D is the rocket body diameter (76mm in this design).

Ensuring the CoM remained ahead of the CoP was essential to flight stability: If the CoM were to fall behind the CoP, aerodynamic forces would overpower control inputs, leading to instability and a loss of control.

With the structural, aerodynamic, and electronic foundations established, the next phase of the project was centered on developing the closed-loop control system responsible for maintaining rocket stability during powered flight. These physical and simulated findings directly informed the PID tuning process and the Kalman filter parameters, ensuring that the control inputs were responsive and accurate during rapidly changing acceleration and vibrations. The following section details the mathematical framework, algorithmic implementation, and practical tuning procedures that governed the TVC's behavior in real time.

3 Control Theory and PID Stabilization

TVC requires real-time orientation correction to counteract disturbances such as wind, manufacturing errors, or uneven thrust. This is achieved through a closed-loop control system, combining an IMU, a Kalman filter, and a dual-axis PID controller that continuously adjusts the rocket’s motor angle to correct pitch and roll deviations.

3.1 Orientation Control using PID

At the core of the TVC system lies a Proportional-Integral-Derivative (PID) controller which stabilizes the rocket by adjusting the motor’s deflection angle in real time. Ideally, the PID controller would oppose external forces acting on the rocket that cause the rocket to deviate away from a normal vertical attitude (0° tilt) . This ensures that the rocket motors’ thrust acts directly parallel to the rocket’s CoM. The controller minimizes the angular error between the desired vertical orientation and the actual tilt, using raw data from the Inertial measurement unit. The PID controller stabilizes pitch and roll independently. Each axis is treated as an independent, separate control loop with its own error data, feedback, and subsequent correction signal to the Servo motors.

The overarching equation representing a PID controller is:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

where:

- $u(t)$: Control output — the commanded servo deflection angle.
- $e(t)$: Measured angular error at time, t , representing the deviation between the measured attitude and the nominal vertical orientation of 0° for pitch and roll.
- K_p, K_i, K_d : The tuned proportional, integral, and derivative gains respectively.

Each term serves a distinct function in stabilizing the rocket. The proportional term (K_p) provides immediate correction to measured angular deviation from the nominal angle. However, if the proportional term is too high, it can induce oscillation. The integral term (K_i) is responsible for removing steady-state error. It does this by summing the error over time (definition of an integral), ensuring that the system returns to its exact setpoint after prolonged deviations. However, a large integral value may cause overshoot or oscillatory drift. The Derivative term (K_d) predicts future error by considering its rate of change, effectively damping the response. It reduces oscillation and improves stability, but excessive derivative gain amplifies measurement noise, especially in systems, like rockets, where IMU data is inherently noisy.

Tuning these gains requires iterative ground and simulated testing to achieve stability without excessive oscillation. Initial values can be estimated using the Ziegler–Nichols heuristic method, which can often lead to excessive PID gains due to its intentional aggressive tuning nature, which is not suitable for a rocket that requires stability and minimal overshoot to avoid excessive oscillation.

This initial theoretical calculation of PID values is followed by empirical adjustment through a static fire test, ensuring that servo commands remain within their actuation limits and that oscillatory behavior was minimized during rapid attitude correction.

Limitations of the PID control model: While the PID control model addresses a way to correct for attitude deviations, it assumes a linearity between the input (servo commands) and outputs (angular deflection). In reality, during powered flight, the relationship between motor deflection and rocket rotation is nonlinear due to aerodynamic effects, a shifting center of mass (from the combustion of propellant), and the thrust curve that is inherent to a solid rocket motor. To mitigate these limitations, the PID controller’s derivative term can be tuned conservatively, and integral windup protection can be applied to avoid overshoot when the system has reached its maximum gimbal correction angle.

3.2 Sensor Fusion using a Kalman Filter

As established, the Control system relies on real-time flight data from sensors. In a perfect world, these sensors would have no noise and translate instantaneous measurements of essential attitude data, but this is not the case. The Gyroscope is accurate for short-term measurements, but measurements drift over time due to inherent limitations in their sensing technology and the way they process data. The Accelerometer provides measurements that are subject to noise under thrust and vibration - two unavoidable fundamentals in rocketry. To ensure that this sensor ‘noise’ is mitigated, a Kalman Filter is implemented to fuse the accelerometer and gyroscope. The Kalman filter works by:

- **Prediction:** Integrating the Gyroscope angular velocity ω to estimate the new orientation.

$$\hat{x}_{k|k-1} = \hat{x}_{k-1} + \omega_k \Delta t$$

- **Update:** Corrects this prediction using accelerometer data, weighted by the filter’s confidence in each sensor:

$$\hat{x}_k = \hat{x}_{k|k-1} + K_k(z_k - H\hat{x}_{k|k-1})$$

where:

- $\hat{x}_k$: Filtered attitude estimate at time step k .
- z_k : Accelerometer-derived measurement of tilt.
- K_k : Kalman gain, which determines how much the new measurement influences the estimate.
- H : Observation model, typically the identity matrix for direct measurement mapping.

The Kalman gain K_k is dynamically updated to minimize the posterior error covariance, allowing the filter to adapt in real-time to changing vibration or thrust conditions. In more simple terms, the Kalman gain (K_k) automatically adjusts in real-time: it constantly recalculates how much to trust to assign to each sensor so the rocket can stay stable even when the engine is vibrating or the thrust, due to the thrust curve, is changing rapidly. The gain is computed as:

$$K_k = \frac{P_{k|k-1}H^T}{HP_{k|k-1}H^T + R}$$

where $P_{k|k-1}$ represents the predicted estimate covariance and R is the measurement noise covariance associated with accelerometer readings.

The outcome is a clean, smooth estimate of the rocket’s orientation that suppresses short-term vibration noise and long-term gyroscopic drift. While the Kalman filter effectively stabilizes the pitch and roll axes, it is important to note that without a magnetometer for an absolute heading reference, the yaw axis is subject to gradual gyroscopic drift. However, for the short duration of a typical model rocket motor burn (approximately 2–5 seconds), this drift is mathematically negligible and does not significantly impact the system’s ability to maintain a vertical ascent. This filtered attitude data is then fed directly into the PID controller, ensuring stable and precise actuation of the gimbal during dynamic phases of flight.

The effectiveness of this filtering approach was validated by comparing simulated IMU data to real flight logs. The Kalman filter consistently reduced instantaneous tilt noise by approximately 70–80%, which in turn improved the responsiveness and accuracy of the PID-driven control loop.

3.3 Stability Constraints and PID gain Tuning Methodology

Designing a closed-loop control system for a thrust-vector-controlled rocket required careful consideration of both control authority and physical response constraints. The PID controller’s ability to stabilize the vehicle depended not only on well-tuned gain values but also on mechanical, aerodynamic, and computational limitations inherent to the component selection for this project. One of the most significant constraints was the servo response speed relative to the rocket’s dynamic instability growth rate. The MG90S servos used in the gimbal system have an average actuation speed of 60° per 0.12 seconds. This means that the control bandwidth is roughly 8-10Hz which is relevant due to the necessity for fast responses to commands from the closed-loop control system. To ensure stability is maximized, the control loop update rate on the Teensy 4.1 Microcontroller was set to 150Hz, significantly higher than the actuator frequency. This allows the controller to detect and respond to deviations before they propagate into larger oscillations that may affect the rocket’s path. The control authority of the system - total torque available from gimbal deflection - was defined by the motor thrust (30.6N) and the gimbal arm length (the perpendicular distance between the thrust vector and the rocket’s centerline). The corrective moment is then defined as:

$$\tau = F_t \cdot r \cdot \sin\theta$$

Where F_T is the instantaneous motor thrust, r is the distance from the thrust line to the rocket’s center of mass, and θ is the gimbal deflection angle. Because the gimbal’s maximum range was $\pm 10^\circ$, the available corrective torque was limited to due the range constraint. Therefore, a limited domain had to be placed on the PID deflection commands to limit the possibility of corrections that would

not be attainable because of the limited deflection angle of the gimbal. This phenomenon is called actuator saturation (where the controller asks for more movement than the motor is physically capable of). The control output of the control system was limited to $\pm 10^\circ$.

3.4 Limits of Control Authority and Tilt Recovery

The ability of a thrust-vector-controlled rocket to maintain or recover stable flight depends fundamentally on the system's control authority - the total corrective torque the TVC mechanism can supply relative to the growth rate of angular disturbances. When the corrective capability of the control mechanism is exceeded, the rocket is no longer able to return to its nominal vertical orientation, resulting in a potential uncontrolled mission failure from a loss of stability. Control authority is limited by a variety of constraints. These limitations are a combination of servo constraints, motor thrust characteristics, aerodynamic forces, structural geometry, and sensor latency. The following subsections explain these limitations in detail and define the theoretical maximum recoverable tilt angle.

3.4.1 Corrective Torque Limitations

The total corrective torque generated by the TVC system is determined by the thrust magnitude of the rocket motor, the offset between the gimbal and the center of mass, and the gimbal's maximum corrective deflection angle. This corrective torque is given by the equation:

$$\tau_{max} = F_T \cdot r \cdot \sin(\theta_{max})$$

Where:

- F_T is the motor thrust.
- r is the perpendicular distance from the thrust line to the center of mass.
- θ_{max} is the maximum gimbal deflection angle.

This brings us to the first mechanical limitation of the corrective system. The control system can only deviate by a fixed degree, meaning that the TVC system has a physically limited torque. As the rocket tilts farther from vertical, a greater corrective torque is required to counteract both gravitational torque and aerodynamic moments. Once the torque of the disturbance exceeds the calculated τ_{max} recovery is an impossibility regardless of how precisely the PID controller is tuned.

3.4.2 Maximum Recoverable Tilt Angle

The maximum recoverable tilt angle defines the largest angular deviation from vertical from which the TVC system can successfully return the rocket to a stable orientation during powered flight. This angle is determined by comparing the maximum corrective torque available from thrust vectoring with the combined disturbance torques acting on the rocket. The main disturbance torques arise from three sources: Aerodynamic restoring or destabilizing moments, which increase with angle

of attack and velocity; Gravitational torque, which increases as the rocket’s center of mass moves laterally away from the thrust axis; Inertial Torque, which depends on the rocket’s angular velocity and moment of inertia. For small angles, using small angle approximations, aerodynamic and gravitational torques scale approximately linearly with tilt angle. As tilt increases, however, these moments grow faster than the available corrective torque due to the fixed gimbal deflection limit. Once the disturbance torque exceeds the maximum available corrective torque defined in Section 3.4.1, recovery is no longer possible, even if thrust remains available. This theorization establishes a theoretical recovery boundary where:

$$\tau_{disturbance}(\theta) \leq \tau_{max}$$

Beyond this boundary, the control system enters a divergent path in which further corrective commands will not provide sufficient angular acceleration to reverse the motion. In practical terms, this means that there exists a finite degree to which the rocket can tilt in which corrective forces can still be applied to stabilize the rocket - otherwise, the flight becomes unrecoverable. Specifically, for this vehicle, simulations and empirical testing indicated that stable recovery was consistently achievable for pitch and roll deviations approximately below 12° , depending on instantaneous thrust and airspeed. This value is slightly greater than the mechanical gimbal limit ($\sim 12^\circ$) due to the fact that recovery relies on angular acceleration over time.

3.4.3 Dynamic Effects and Control Latency

In addition to static torque limitations, tilt recovery is constrained by dynamic effects related to control latency, actuator response speed, and sensor update rates. Even when sufficient corrective torque exists in theory, delays in these other factors can allow angular deviations to grow beyond recoverable limits before corrective torque can be applied. In a perfect world, this delay would not exist, but even milliseconds of control delay can cause the system to over or undercorrect. The MG90S servos used in this specific TVC system exhibit a finite response time of approximately 0.12 seconds per 60° of motion. This inherently poses a limitation to how quickly the thrust vector can be redirected in response to sudden disturbances. For example, if the rocket experiences a rapid tip-off event immediately after launch, as often happens with light-weight aerial vehicles, the servo response may lag behind the disturbance, allowing angular velocity to increase faster than the system can react. In a similar way, sensor latency and filtering introduce additional marginal delay. Although the IMU provides data frequently, the Kalman filter smooths rapid jumps in sensor data to reduce noise. This introduces a small but unavoidable phase lag in attitude estimation. While this tradeoff improves stability under vibration, it reduces responsiveness to extremely fast disturbances. These combined effects impose a dynamic recovery limit, where recovery fails not because torque is insufficient, but because corrective movement occurs too late. To mitigate this risk, conservative PID gains can be selected, prioritizing stability and predictability over aggressive corrections. Additionally, the control loop frequency should be set significantly higher than the actuator bandwidth to minimize computational delay and ensure that corrective commands are issued as early as possible. To conclude, these constraints define a very specific, calculatable domain

of controllability for the thrust-vector-controlled rocket, within which stable flight and recovery are a possibility. Understanding this domain was essential in both the mechanical design of the gimbal system and the tuning of the closed-loop control algorithm.

4 Flight Software and Autonomous State Transitions

The Flight software serves as the layer that governs when the control system is enabled, how sensor data is interpreted, and how the system transitions between distinct phases of flight. While the thrust vector control system defines how corrections actions are computed, the flight software determines when such corrections are physically necessary and safe to apply. For this project, an autonomous state-machine-based architecture was implemented to ensure data-backed behavior, robustness that protects against sensor noise, and safe transitions between powered and unpowered flight. The software executes on a Teensy 4.1 microcontroller as a high-frequency loop ($\sim 200\text{Hz}$). Within each iteration, sensor data is sampled, filtered, and evaluated to determine the rocket’s current flight state. State transitions occur only when clearly defined physical conditions are satisfied, recurring the likelihood of false detections and unintended control activation. This approach allows the system to remain predictable under both nominal and non-nominal conditions while maintaining a clear separation between guidance, control, and recovery logic.

4.1 Autonomous Staging and Flight Phase Identification

At the core of the flight software lies the autonomous staging system, responsible for identifying the rocket’s current phase of flight based on real-time sensor data. This system evaluates accelerometer, gyroscope, and derived kinematic quantities to determine whether the rocket is idle, undergoing powered ascent, coasting, descending, or has completed flight. Each flight phase corresponds to a unique control context, ensuring that thrust vector corrections are applied only when thrust is available and that recovery logic is activated at the appropriate moment. Rather than relying on a single sensor or threshold, the staging logic evaluates multiple indicators simultaneously. This multi-parameter approach improves robustness against noise, vibration, and transient disturbances, particularly during high-dynamic events such as launch and motor burnout. The overall staging logic follows the structure illustrated in Figure A1, which outlines the transitions between discrete flight states.

4.2 State Machine Logic and Execution Flow

The flight software is structured as a state machine operating within a continuous control loop. During each iteration of the loop, current sensor measurements are processed and compared against state-dependent transition criteria. The system begins in a pad idle state, during which the rocket is stationary and awaiting launch. In this state, sensor calibration is completed, baseline offsets are established, and all actuators are held in neutral positions to prevent unintended motion.

Upon detection of launch conditions, the system transitions into the powered flight state. In this phase, thrust vector control is fully enabled, and the PID controller actively stabilizes pitch and

roll using Kalman-filtered attitude estimates. The powered flight state persists until motor burnout and upward velocity decay are detected, at which point the software transitions into an apogee detection state. This intermediate state allows the system to distinguish between transient velocity fluctuations and the true apex of the trajectory.

Once apogee is confirmed, the system enters the descent state. During descent, active thrust vector control is disabled, as thrust is no longer available to generate corrective torque. The final transition occurs after landing is detected, at which point the system enters a recovery or post-landing state where data logging is finalized and active control subsystems are powered down.

This state-based execution model ensures that control logic is always context-aware and prevents undesired actuation when it is not required or will not have a meaningful effect on controlling the rocket's trajectory.

4.3 Sensor Inputs, Calibration, and Filtering

Reliable autonomous decision-making depends on accurate and stable sensor data. The rocket employs a six-degree-of-freedom MPU-6050 inertial measurement unit, providing three-axis accelerometer and gyroscope measurements. While these sensors offer high sampling rates and low latency, their raw outputs are susceptible to bias, drift, and vibration-induced noise. Gyroscopes measure angular velocity and therefore accumulate integration error over time, leading to gradual drift in estimated orientation. Accelerometers, while capable of providing absolute tilt references, are sensitive to high-frequency vibration and non-gravitational accelerations during powered flight. To address these limitations, sensor calibration routines can be performed prior to launch to remove static offsets and real time filtering is applied during flight. Filtered sensor data is then passed to the Kalman filter described in Section 3.2, producing a smooth and reliable estimate of the rocket's orientation. This filtered attitude estimate is used both for control input during powered flight and for flight state determination throughout the mission.

4.4 Launch Detection Criteria

Accurate launch detection is critical for enabling thrust vector control at the correct moment. Premature activation can lead to unnecessary servo actuation on the launch rail, while delayed activation reduces the system's ability to counter early disturbances. To balance responsiveness and robustness, launch detection is based on multiple simultaneous conditions rather than a single acceleration threshold.

The transition from pad idle to powered flight is triggered only when sustained axial acceleration exceeding a predefined threshold is detected, accompanied by a consistent increase in vertical velocity over a short time window. Additionally, a minimum arming time must elapse to prevent accidental activation during handling, and gyroscope noise must remain below a defined threshold to ensure a stable sensor environment. Together, these conditions significantly reduce the probability of false positives while ensuring that thrust vector control engages directly after liftoff.